

Program 7

Deploy a Web Application to Kubernetes using Minikube

The program demonstrates how to deploy a simple web application (Nginx) on a local Kubernetes cluster using Minikube. The aim is to understand the process of creating a Deployment, exposing it through a Service and accessing the web application running inside Kubernetes pods.

Project Structure:

program_7/

—>nginx-deployment.yaml

Procedure:

Step 1: Start Minikube and Check the status of Minikube

minikube start

minikube status

Step 2: Create nginx-deployment.yaml file with the following code

nano nginx-deployment.yaml

```
GNU nano 7.2 nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 3: Apply the nginx-deployment.yaml file and Check status

```
kubectl apply -f nginx-deployment.yaml
```

```
kubectl get pods
```

```
kubectl get deployments
```

Step 4: Expose the deployment as service, Check services and Acces the web app

```
kubect expose deployment nginx-deployment - -type=NodePort - -port=80
```

```
kubectl get services
```

```
minikube service nginx-deployment
```

```
Dec 12 10:49
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8: ~/Devopslab/program7

! Failing to connect to https://registry.k8s.io/ from both inside the minikube container and host machine
! To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
! Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
! Verifying Kubernetes components...
!   Using image gcr.io/k8s-minikube/storage-provisioner:v5
!   Using image docker.io/kubernetes/dashboard:v2.7.0
!   Using image docker.io/kubernetes/metrics-scraper:v1.0.8
! Some dashboard features require the metrics-server addon. To enable all features please run:

minikube addons enable metrics-server

! Enabled addons: default-storageclass, storage-provisioner, dashboard
! Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8: ~/Devopslab/program7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8: ~/Devopslab/program7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created

jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8: ~/Devopslab/program7$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
kubia                                0/1     ImagePullBackOff    0           3d18h
nginx-deployment-d8fcc84f-bv86r      0/1     ContainerCreating    0           7s
nginx-deployment-d8fcc84f-jp26w      0/1     ContainerCreating    0           7s
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8: ~/Devopslab/program7$ kubectl get deployments
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment                    0/2     2             0           13s
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8: ~/Devopslab/program7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed

jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8: ~/Devopslab/program7$ kubectl get services
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes                          ClusterIP            10.96.0.1        <none>            443/TCP           17d
kubia                               NodePort             10.98.248.134    <none>            8080:30597/TCP    3d18h
kubia-http                          LoadBalancer        10.110.114.156   <pending>         8080:30528/TCP    2d13h
nginx-deployment                    NodePort             10.108.214.172   <none>            80:30949/TCP      10s
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8: ~/Devopslab/program7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30949